Merging Staff and Site Datasets

https://gemini.google.com/app/9b30293dfa06aba8

User prompt: I have two datasets that I need to work with. This codebook is also here. First, how many rows and what is a row in the adequately joined data?

Response: To combine `staff_quarters.csv` and `site_weeks.csv` properly, you must first aggregate one or both tables so they share a common unit of observation (grain). A direct, unaggregated join on `site_id` and `quarter` creates a many-to-many Cartesian product of **41,793 rows** (each employee row duplicated ~13 times for every week in the quarter), which artificially inflates sample sizes and skews metrics.

Depending on your analytical goal, there are three valid choices for an adequately joined dataset:

1. Employee–Quarter Level (Recommended for HR & Turnover Analysis)

- **Row Count: 3,215 rows**
- **What a Row Is: One employee in one calendar quarter**, enriched with summary statistics of their site's quarterly performance.
- **Aggregation Method:** Aggregate `site_weeks.csv` to the site-quarter level (calculating quarterly averages or totals for weekly metrics like client satisfaction or missed visits), then join onto `staff_quarters.csv` on `site_id` and `quarter`.
- **Best Used For:** Analyzing employee turnover (`left_org`), engagement, and overtime relative to site conditions.

2. Site–Week Level (Recommended for Operational Analysis)

- **Row Count: 832 rows** ($8 \text{ sites} \times 104 \text{ weeks}$)
- **What a Row Is: One site in one specific week**, enriched with the site's quarterly staffing metrics.
- **Aggregation Method:** Aggregate `staff_quarters.csv` to the site-quarter level (calculating quarterly averages for pay, tenure, headcount, or turnover rate), then join onto `site_weeks.csv` on `site_id` and `quarter`.
- **Best Used For:** Analyzing weekly site performance (intake wait days, documentation errors, client satisfaction) relative to staffing characteristics.

3. Site–Quarter Level (Recommended for Program Evaluation)

- **Row Count: 64 rows** ($8 \text{ sites} \times 8 \text{ quarters}$)
- **What a Row Is: One site in one calendar quarter**, summarizing both staff and operational performance.

- **Aggregation Method:** Aggregate both `staff_quarters.csv` and `site_weeks.csv` to the `site_id` + `quarter` level before merging.
- **Best Used For:** Evaluating overall site trends and testing the macro-impact of SmartShift across sites over time.

User prompt: This site week level is interesting. Show me the week of the quarter on x, cases closed on y, and facet by quarter.



Here is the faceted plot showing **Week of Quarter** versus **Cases Closed**, faceted across all 8 calendar quarters (2024Q1–2025Q4).

Key Pattern & Notable Finding

- **Site S2 Quarterly End-of-Quarter Batch Closing:** Across every single quarter, **Site S2** maintains a baseline of ~30–40 cases closed per week during weeks 1 through 12, followed by a massive spike in the final week of the quarter (closing **518 to 626 cases** in a single week).
- **Other Sites (S1, S3–S8):** All other sites maintain relatively steady weekly case closures throughout each quarter (ranging between 30 and 100 closed cases per week).

This pattern in S2 indicates a procedural administrative practice where case closures are accumulated and logged in bulk at the end of each quarter rather than recorded continuously week-by-week.

Would you like to investigate whether this quarterly batch-closing behavior at Site S2 correlates with higher documentation error rates or longer intake wait times in those final weeks?
